

Sruthi Satyavarapu

AI/ML Engineer | LLM Systems | RAG & Agentic AI | MLOps

sruthisatya27@gmail.com • +1 510-365-2100 • San Jose, CA

[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

PROFESSIONAL SUMMARY

AI/ML Engineer with 7+ years of end-to-end experience from architecting secure backend systems to deploying production LLM applications, RAG pipelines, and real-time ML services at scale. Skilled across the full AI lifecycle: data engineering, model fine-tuning, evaluation (RAGAS, MRR, Recall@K), and MLOps on AWS SageMaker, Docker, and Kubernetes. Delivered proven results across PNC Financial, Tata Consultancy Services, and University of the Pacific research labs.

TECHNICAL SKILLS

- LLM & GenAI:** LLM Fine-Tuning (LoRA/QLoRA), Prompt Engineering, RAG, LangChain, LlamaIndex, Hugging Face, OpenAI API, AWS Bedrock
- ML / Deep Learning:** PyTorch, Scikit-learn, NLP, RAGAS, Recall@K, MRR, Time-Series Analysis, Anomaly Detection, XGBoost
- Vector Databases:** ChromaDB, Pinecone, FAISS, Weaviate embedding pipelines, similarity search, hybrid retrieval
- Cloud Platforms:** AWS (SageMaker, EC2, S3, Lambda, Bedrock), Docker, Kubernetes
- Languages:** Python, SQL, JavaScript, TypeScript, C++, PHP, Java
- Agentic AI:** AI Agent Design, Tool-Calling Agents, LangGraph, AutoGen, Multi-step Reasoning, Autonomous Pipelines
- MLOps & Cloud:** AWS SageMaker, MLflow, Docker, Kubernetes, Apache Airflow, CI/CD, Model Monitoring, Model Drift Detection
- Backend & APIs:** FastAPI, Flask, Spring Boot, RESTful Microservices, ETL Pipelines, Yii Framework
- Data & Visualization:** Pandas, Matplotlib, Seaborn, Tableau, d3.js, Apache Spark basics
- Security & Auth:** OWASP, OAuth2, JWT, RBAC secure API and multi-user system design

EDUCATION

M.S. Computer Science | University of the Pacific, Stockton, CA

Aug 2024 – May 2026 | GPA: 4.0

M.S. Technology (Medical Software) | Manipal Institute of Technology, India

Aug 2014 – May 2016 | GPA: 3.7

RESEARCH EXPERIENCE

Research Assistant | RAG & LLM Systems | **University of the Pacific**

Mar 2025 – Present • Stockton, CA

- Designed modular RAG architecture bridging benchmarking frameworks (FlashRAG) and production frameworks (LangChain, LlamaIndex), enabling configuration-driven experimentation without architectural changes applying LoRA-based fine-tuning for domain-adaptive retrieval.
- Built agentic query pipeline using ReactJS, FastAPI, and ChromaDB with query rewriting, cross-encoder re-ranking, session-aware interaction, and RBAC access control for multi-user environments; integrated persistent storage and latency benchmarking for reproducible cross-domain RAG evaluation.
- Evaluated retrieval performance across biomedical and Sanskrit philosophical corpora using Recall@K, MRR, and RAGAS demonstrated that component selection drives significant domain-specific performance variance.
- Paper under review IEEE BigData Service: 'A Modular Architecture for Domain-Adaptive Retrieval-Augmented Generation Systems'

Research Assistant | ML for IoT & Health Systems | **University of the Pacific** *Jun 2025 – Dec 2025 • Stockton, CA*

- Developed anomaly detection models using PyTorch and Scikit-learn on wearable sensor data streams for IoT-based health monitoring, improving detection accuracy by 28% over baseline.
- Engineered time-series feature extraction pipelines for environmental and motion-based sensor datasets, enhancing predictive model performance across multiple sensor modalities.
- Built Flask-based RESTful microservices integrated with React dashboards to deliver real-time sensor insights and ML model outputs to end users; containerized with Docker for reproducible deployment.

INDUSTRY EXPERIENCE

AI/ML Engineer Intern | **PNC Financial Services, USA**

Jan 2024 – Jul 2024 • Remote, USA

- Built end-to-end ML pipelines on AWS SageMaker and Apache Airflow for financial risk scoring systems, improving processing efficiency by 30% and reducing model serving latency.
- Developed NLP-based fraud detection models leveraging fine-tuned LLMs and ensemble ML techniques, reducing detection latency by 15% and improving real-time risk scoring accuracy.
- Designed and deployed scalable ML inference services via Java Spring Boot microservices, Docker, and Kubernetes across cloud environments, ensuring high availability and seamless enterprise integration.

Senior Software Developer | ML & Backend | Tata Consultancy Services Jul 2021 – Apr 2023 • Hyderabad, India

- Developed and deployed ML models for predictive analytics on the Mars Petcare platform, improving demand forecasting accuracy by 25% using gradient boosting and time-series modeling.
- Implemented OWASP-compliant secure backend systems for enterprise applications, improving performance by 20% and reducing security vulnerabilities by 35%.
- Deployed and maintained scalable cloud infrastructure on AWS (EC2, S3, Lambda) supporting high availability for production ML services.

Web Developer | Hyper Interact Services

Feb 2020 – May 2021 • India

- Developed and managed full-stack web applications tailored for SEM (Search Engine Marketing) campaigns, driving measurable improvement in user engagement.
- Created and optimized landing pages to enhance conversion rates, applying A/B testing and performance monitoring techniques.

Technologies: JavaScript, HTML/CSS, PHP, MySQL

Senior Software Engineer | Foray Software Private Limited

Jul 2019 – Oct 2019 • India

- Enhanced front-end inventory search functionality for Mars employees, delivering 30% faster search experience and 15% increase in task efficiency.
- Partnered with cross-functional teams to integrate features and ensure seamless system functionality, improving overall user satisfaction scores.

Technologies: Drupal, JavaScript, MySQL

Associate Analyst | PurpleTalk

Feb 2017 – Jun 2019 • India

- Developed peer-to-peer digital services features for Android application, enabling seamless user-to-user transactions. Canya App.
- Contributed to development and unit testing of a social networking alarm app; received Best Performance Award for simultaneous management of multiple projects. Cuckuu App.
- Developed and maintained the payment and invoice module for an enterprise attendance management system; received Best Employee Award for outstanding contributions. SILA Attendance System.
- Implemented key modules for this user engagement application improving feature set and stability. Upshot.

Technologies: Yii Framework, MongoDB, JavaScript, jQuery

Research Associate | Excelra

Jun 2016 – Dec 2016 • India

- Pathway Enrichment Tool: Developed an automated visualization tool to map disease associations and biological pathways for a given target gene ID, improving workflow efficiency by 25% and accelerating bioinformatics analysis.

Technologies: PHP, JavaScript, jQuery, Ajax, HTML, CSS, d3.js

KEY PROJECTS

AI Companion for Seniors [Ongoing] | AWS Bedrock, RAG, MongoDB, FastAPI, React, Python

- Voice-enabled multimodal AI health platform using AWS Bedrock and agentic RAG pipeline supports medical data extraction, medication tracking, anomaly detection on vitals, and real-time caregiver alert dashboard with multimodal input.

Strawberry Fruit Ripeness Detection | Python, Scikit-learn, Pandas

- Built a supervised ML model (KNN + SVM ensemble) achieving 92% accuracy for automated strawberry ripeness classification; authored a white paper detailing methodology, feature engineering, and findings.

Big Data Analysis — Weather & Music Trends | Python, Matplotlib, Seaborn, Tableau

- Performed exploratory data analysis across 100+ cities correlating weather patterns with music chart trends and genre preferences; presented insights via interactive Tableau dashboard.

Adidas Sales Dashboard | Python, Tableau

- Developed an interactive Tableau dashboard analyzing 2023–2024 Adidas sales data, uncovering regional performance trends and product-level insights to support data-driven business decisions.

Desert Survival Game | C++

- Designed and implemented a grid-based world map system in C++ where players navigate locations, collect resources, and craft tools to survive demonstrating strong OOP and systems programming fundamentals.